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Federated Traffic-Flow Prediction with Communication-Cost

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Problem Description

Centralised traffic-flow prediction requires raw traffic data from multiple intersections to be sent to a central server, increasing communication costs and raising privacy concerns. This project proposes a Federated Learning (FL) approach in which simulated intersections train models locally and share only model updates using Federated Averaging (FedAvg).

Using a public traffic dataset distributed across 3–5 simulated clients, the project compares centralised and federated learning under IID and non-IID data distributions. Performance is evaluated using MAE, RMSE, communication bytes per round, and model convergence across three random seeds to analyse the trade-off between prediction accuracy and communication overhead.

Background

Traffic-flow prediction plays an important role in smart cities by helping analyse traffic patterns and improve traffic management. Traditional centralised machine learning requires traffic data from multiple intersections to be transferred to a central server, which can increase network bandwidth consumption and raise privacy concerns. Federated Learning (FL) provides an alternative approach in which each simulated intersection trains a model locally and shares only model updates with a central server. Using Federated Averaging (FedAvg), the server combines these updates to create a global prediction model. This project compares centralised and federated learning using IID and non-IID traffic data to evaluate prediction accuracy, communication costs, and model convergence.

Literature & RFC Review

Existing Studies, Proposals and Standards

McMahan et al. (2017) introduced Federated Averaging (FedAvg), allowing multiple clients to train models locally and share model updates instead of raw data. Existing research on traffic forecasting using Federated Learning demonstrates its potential while highlighting challenges related to non-IID data and communication overhead. These studies motivate our approach of comparing centralised and federated traffic prediction using IID and non-IID data distributions. The project evaluates prediction accuracy using MAE and RMSE, measures communication costs, and analyses model convergence across three random seeds.

Proposed Solution

The proposed solution develops a Federated Learning simulator for traffic-flow prediction using a public traffic dataset distributed across 3–5 simulated intersections. Each intersection trains a compact forecasting model locally without sharing its raw traffic data. A central server collects the model updates and combines them using Federated Averaging (FedAvg) to produce a global model.

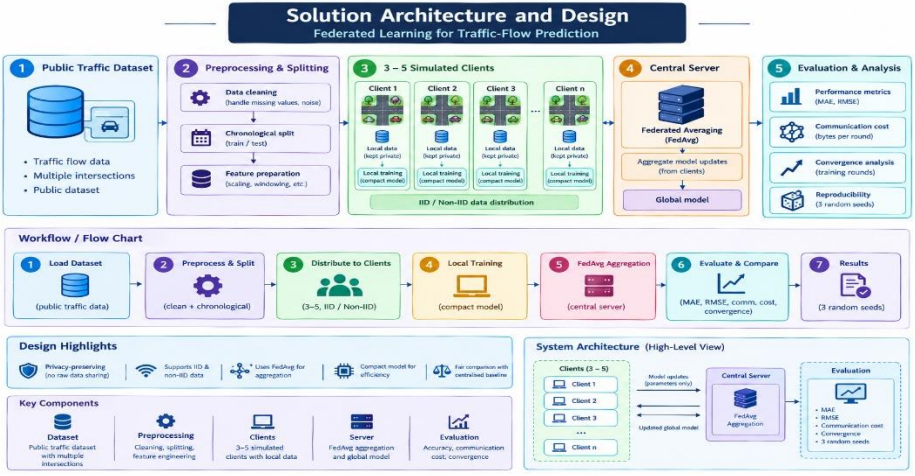
The system compares federated learning with a centralised learning baseline under IID and non-IID data distributions. Model performance is evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), while communication overhead is measured in bytes transmitted during each training round. Model convergence is also analysed to understand how performance changes over successive rounds.

Optimization Proposed

The proposed system incorporates several optimizations to improve traffic-flow prediction and reduce communication overhead. Federated Learning keeps raw traffic data local to simulated intersections, while Federated Averaging (FedAvg) combines locally trained model updates to create a global model. The system compares IID and non-IID data distributions to understand how differences in traffic patterns affect prediction performance. A compact forecasting model helps manage computational and communication requirements. Chronological train-test splitting prevents data leakage and ensures a fair evaluation. Model performance is measured using MAE and RMSE, while communication bytes per training round and model convergence are tracked to assess efficiency. Finally, experiments with three random seeds improve the reliability and reproducibility of the results, enabling a systematic comparison between centralised and federated learning.

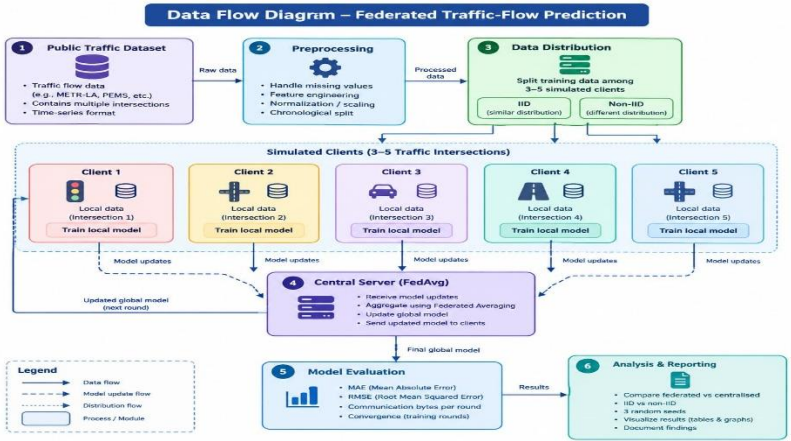
Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Data Flow Diagram

Overview of the architecture. Include the flow chart diagram created earlier. Visual: Diagrams or architecture visuals.



Deployment Timeline

The project will be developed in several milestones, beginning with the selection and preprocessing of a public traffic dataset, followed by a chronological training and testing split. Next, the dataset will be distributed among 3–5 simulated clients using IID and non-IID data distributions. A compact traffic-flow forecasting model will then be developed, along with a centralised learning baseline for comparison. In the federated learning phase, each client will train the model locally and send only model updates to the central server, where Federated Averaging (FedAvg) will be used to generate an updated global model. The system will then be evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), communication bytes per training round, and model convergence.

Bibliography I

1 1. McMahan, B., Moore, E., Ramage, D., Hampson, S., & Agüera y Arcas, B. (2017).Communication-Efficient Learning of Deep Networks from Decentralized Data. Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), 1273–1282.

2 Li, T., Sahu, A. K., Talwalkar, A., & Smith, V. (2020). Federated Learning: Challenges, Methods, and Future Directions. IEEE Signal Processing Magazine, 37(3), 50–60.

Thank You for Your Attention!